

Project Management, Risk Management and Change Management





Project Management

- **Definition**

- Project Management

- is the process of planning, organizing and delineating responsibility for the completion of an organizations' specific information technology (IT) goals.

- Project management includes overseeing projects for

- software development,
 - hardware installations,
 - network upgrades,
 - cloud computing and virtualization rollouts,
 - business analytics and data management projects.



Project Management

- **Managing Project**

- Project Management Life Cycle

- Initiation – the project goal, need or problem is identified. The Project manager is assigned to the project and the Project charter is created.
- Planning – the project manager and the project team work together to plan all the needed steps to reach a successful project conclusion. The project planning processes are iterative in nature, and it's expected that planning will happen often throughout the project.
- Execution – once the Project plan has been created, the project team goes about executing the project plan to create the deliverables of the project. The project can shift to project planning as needed throughout project execution.
- Monitoring and controlling – as the project is being executed by the project team, the project manager monitors and controls the work for time, cost, scope, quality, risk, and other factors of the project. Monitoring and controlling is also an ongoing process to ensure that the project addresses its targets for each project objective.
- Closing – at the end of each phase and at the end of the entire project, project closure happens to ensure that all of the work has been completed, is approved, and ultimately transferred ownership from the project team to operations.



Project Management

- **Managing the Project Knowledge Areas**

- The ten project management knowledge areas are:

- Project scope management: the project scope is defined, documented, and approved. The project scope is protected from unauthorized changes, edited with approved changes, and validated by the Project stakeholders for project acceptance.
- Project schedule management: the project schedule is defined first by the working hours of the project, any project milestones, and ultimately a project deadline.
- Project costs management: the costs of the project are estimated so that a budget for the project can be assigned.
- Project quality management: what constitutes quality in the project is defined in specific metrics and agreed upon among the stakeholders as early in the project as possible. Quality assurance programs and policies direct the project work, while quality control inspects the project work to confirm that quality has been ascertained in the work.
- Project communications management: stakeholders will need information from the project manager will need to provide information to the project manager throughout the project life cycle. This knowledge area create a communications management plan that address who will need what information, when the information is needed, and the best modality for the communications.



Project Management

- **Managing the Project Knowledge Areas**

- The ten project management knowledge areas are:

- Project risk management: risks are situations, events, conditions that can threaten, and sometimes benefit, the objectives of the IT project. Risks must be identified, analyzed, and a response created for the risk event. The probability and impact of each risk event is evaluated to create a risk score to justify the costs needed to manage the risk event.
- Project procurement management: should the project need to purchase goods or services, a formal process for procurement will need to be created. The plan should address the project's selection of contract type, administration of the contract, purchasing audits, and contract closeout. Many project managers do not manage procurement but defer to the organization's centralized procurement or purchasing department and processes.
- Project stakeholder management: stakeholders are anyone that has a vested interest in the project. Stakeholder management is the identification, inclusion, and communication with the groups of project stakeholders. It manages the anxieties and concerns the stakeholders may have about the project work.
- Project integration management: this special knowledge area is the coordination of the events in all the other knowledge areas. How well the project manager performs in one knowledge directly affects the performance of the other knowledge areas. Project integration management examines the interactions and contingencies among the knowledge areas to ensure that the project is adequately planned, executed, controlled and closed.



Risk Management

- **Definition**
- IT risk management is defined as the policies, procedures, and technology an organization adopts in order to reduce the threats, vulnerabilities, and consequences that could arise if data is not protected.
- IT risk management examines this classic equation for risk:

Threat x Vulnerability x Consequence



Risk Management

- **Risk Equation**
- **Threat** is inherent in IT risk management, and most organizations assume that their vendors present at least some level of threat.
- **Vulnerability** comprises the gaps in a protection program. Let's say you have a sensitive document and you put it in a safe, in a locked building, protected by guards; you likely feel confident that the document isn't vulnerable to many threats. Now, if that same document was online in an open network where your organization stores all its data, it's easy to understand how the safety of the document is compromised. The moral of this story is to understand not only what vulnerabilities are in your protection program, but also how those vulnerabilities can be exploited. Once you parse through this information, you'll gain a clearer idea of how to address your overall risk
- **Consequence** represents the harm caused to an organization by a cyberattack. An important element to consider here is the value of the information you're trying to protect — something which can vary tremendously. For example, intellectual property data or pricing information may be of value to your organization. But data, such as personally identifiable information (PII), can also hold value because of the legal requirements to protect it. When determining risk, it's important to ask what might happen if that data is compromised.



Risk Management

➤ Steps to Risk Management

- **Identify the Risk** You can't prepare for risk without first figuring out, to the best of your ability, where and when it might arise. Therefore, both manager and team must be alert to uncovering and recognizing any risks, then detailing them by explaining how they might impact the project and outcomes.
- **Analyze the Risk** Once you've identified risk, you then must analyze it and discern if it's big, small or minimal in its impact. Also, what would be the impact for each of the risks. Study the risk and how it might influence the project in various ways. You'll add these findings to your risk assessment.
- **Evaluate and Rank the Risk** Once you evaluate the impact of risks and prioritize them, you can begin to develop strategies to control them. This is done by understanding what the risk can do to the project, which is determining the likelihood of it occurring and the magnitude of its impact. Then you can say that the risk must be addressed or can be ignored without faulting the overall project. Again, these rankings would be added to your risk assessment.
- **Respond to the Risk** After all this, if the risk becomes an actual issue, then you're no longer in the theoretical realm. It's time for action. This is what's called risk response planning in which you take your high-priority risks and decide how to treat them or modify them, so they place as lower priority. Risk mitigation strategies apply here, as well as preventive and contingency plans. Add these approaches to your risk assessment.
- **Monitor and Review the Risk** Once you act, you must track and review the progress on mitigating the risk. Use your risk assessment to track and monitor how your team is dealing with the risk to make sure that nothing is left out or forgotten.



Risk Management

- **Risk Management Strategies**

- **Apply Safeguards**

- This is an avoidance strategy, where the company decides to avoid risk at all costs and focuses a great deal of resources to that end. If you can avoid the risk, then it is no longer a threat to the project. However, there is a downside to this. If you avoid the risk, you also avoid the associated potential of its return and opportunity. So, it's a decision not to take lightly.

- **Transfer the Risk**

- This is a transference strategy, when the company transfers the risk to another entity. This redistribution can be onto the company members, some outsource entity or an insurance policy

- **Reduce the Impact**

- This is a mitigation strategy, where the company works to reduce the impact of the risk through methodology, teams or whatever resources are at its disposal. It can involve small changes, but always must come by process and a plan.

- **Accept the Risk**

- This is an acceptance strategy, where you know there is risk and accept that, so when and if it occurs you can deal with it then and there. This is sometimes unavoidable, but manageable if you have followed the steps in your project risk assessment template.



Risk Management

- **Best Practices for Risk Management**
- **Evaluate Early & Often:** There's no better time to start on the risk management process than now, so begin early. Remember it is a process and so it will continue throughout the project. Then continue monitoring all the time. Risk never sleeps.
- **Lead from the Top:** Good leadership is many things. One aspect is developing a risk culture at the organization. That means valuing input from everyone, believing in the importance of acknowledging risk and keeping a positive attitude about responding.
- **Communications:** Having a clear channel to communicate risk throughout the organization is paramount to identifying and responding quickly and effectively to risk.
- **Strong Policies:** If there is not already a process and plan to deal with risk, you're always going to be one step behind. This is again why a project risk assessment is key, but so is understanding roles and responsibilities for everyone on the Project team, having a continuity plan, etc.
- **Involve Stakeholders:** A great resource that is often overlooked are the Project stakeholders, who have a unique perspective and can provide insight into areas where risk might arise. So, involve them throughout the process, from asking for their participation with the risk assessment template and over the whole course of the project.
- **Get Signoffs:** At every stage of your risk management, get people to sign-off on the strategy, which includes the stakeholders.



Change Management

- **Definition**

- IT Change Management is an IT Service Management (ITSM) process that makes it easier for your organization to roll out change requests to your IT infrastructure. How exactly? It helps your organization to request, prioritize, authorize, approve, schedule and implement any changes. Whether the change is complex or simple, a good **IT Change Management process** helps you to control risk and keep disruptions to your services to a minimum



Change Management

- **Benefits of IT Change Management**

- Implement changes more quickly
 - Track the progress of changes to your IT infrastructure
 - Make your IT Change Management process more transparent, improving communication with stakeholders
 - Easily trace the roll-out of any changes if something goes wrong
 - Improve cost estimate for any proposed changes
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Change Management



- **What kind of changes are there?**
- In IT Change Management there are three distinct classes of change: standard changes, normal changes and emergency changes. Each class of changes is managed in a different way:
 - **Standard changes.** These are changes to a service or to the IT infrastructure. Here, the implementation process and the risks are known upfront. These changes are managed according to policies that an IT organization already has in place. An example of a standard change: installing a new printer, or software on a laptop.
 - **Normal changes.** These changes must go through a change process before they can be approved and implemented. If they are high-risk, a change advisory board* decides whether they will be implemented. A normal change could be moving a new development project into production or adding a new server.
 - **Emergency changes.** These are changes that must be performed as soon as possible. An example of an emergency change: having to fix a security breach that requires a patch to many workstations.



Change Management

- **What's the difference between a change, an incident and a problem?**
- So, if a change is an activity to improve the IT infrastructure of your organization, while keeping disruptions to a minimum. Where do incidents and problems fit in?
- Let's look at the definitions. An incident is a single event in which one of your organization's (internal) services are not performing as you'd like. An example? Think of a laptop that won't boot properly. On the other hand, a problem is used to register recurring interruptions (or incidents) to your organization's IT structure.
- An example. Say you've just had lunch and notice that there are over twenty incidents of a printer on the second floor not working. Together, these incidents form a problem about a defect printer. Finally, you could start a change to replace the printer with a new one.



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